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Compressibility Effects on a Minimum Pressure Gradient Nose

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Introduction

In ref. (1) a technique to develop nose profiles with minimal pressure gradients in incompressible flow was described. The motivation for developing such shapes was the hope that the forebody boundary layer separation at angle of attack into a free vortex pair could be delayed or avoided entirely. Such vortex pairs are known to have severe adverse interaction with a vehicle's tail fins.

One application of these forebody profiles is to university sounding rockets. Many of these have burnout Mach numbers in the high subsonic regime. This raises the possibility that local supersonic flow over the nose near the minimum pressure region will occur, followed by abrupt downstream deceleration in a normal shock. This, of course, would tend to enhance boundary layer separation, even though minimizing the surface pressure would postpone shock-induced separation to a higher Mach number than would otherwise be the case.

Karman-Tsien Compressibility Correction

Reference (1) showed that it was possible to find a nose profile whose minimum incompressible pressure coefficient Cp was about half of that for a cone-cylinder of the same fineness ratio. Figure 1 below shows a computed incompressible pressure coefficient distribution for an L/D = 6, 36" long, nose profile with the least possible pressure coefficient (~ – 0.027).

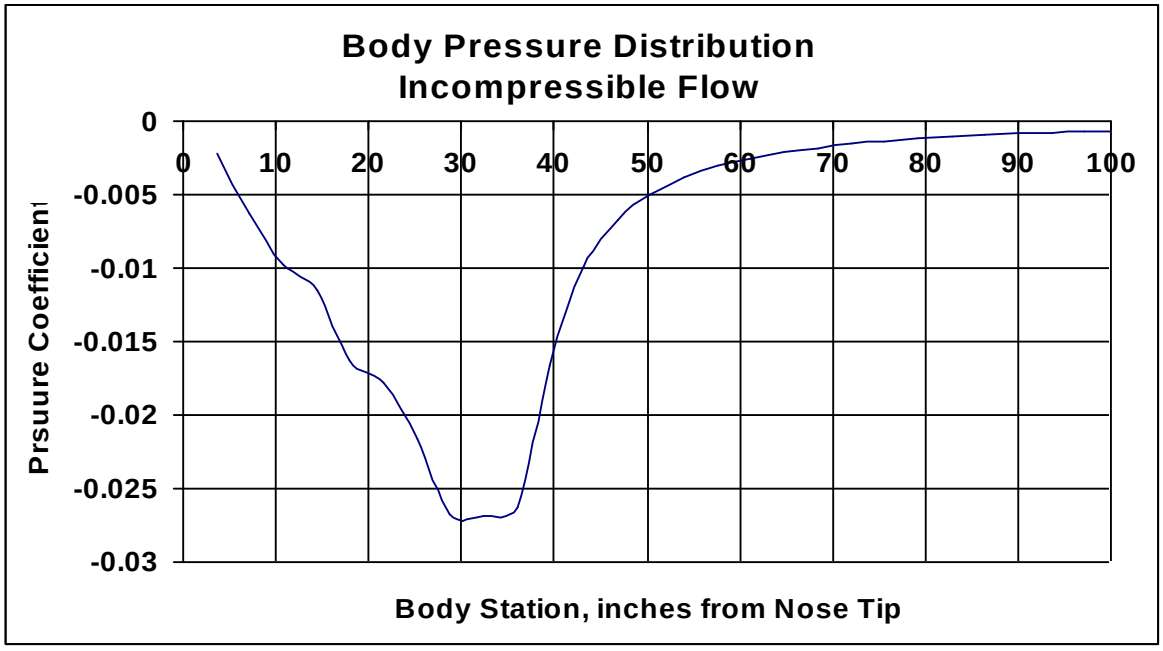


Fig. 1 Optimum Incompressible Pressure Coefficient for L/D = 6

Now, consider the same profile in compressible, subcritical flow. Subcritical means the free stream Mach number is sufficiently small no local regions of supersonic flow can occur. While there several ways of obtaining compressible flows from incompressible results, the technique used here is that due to von Karman and Tsien, ref. (2). The Karman-Tsien formula is

$$Cp(M_\infty) = Cp(0) \,/\, \{\beta + \tfrac{1}{2}\, Cp(0)\, [\, 1 - \beta\,]\}, \text{ where} \qquad (1)$$

$$\beta = \sqrt{1 - M_\infty^2}$$

This can be used to estimate the pressure coefficient at $M_\infty = 0.7$ as shown below in Fig. 2:

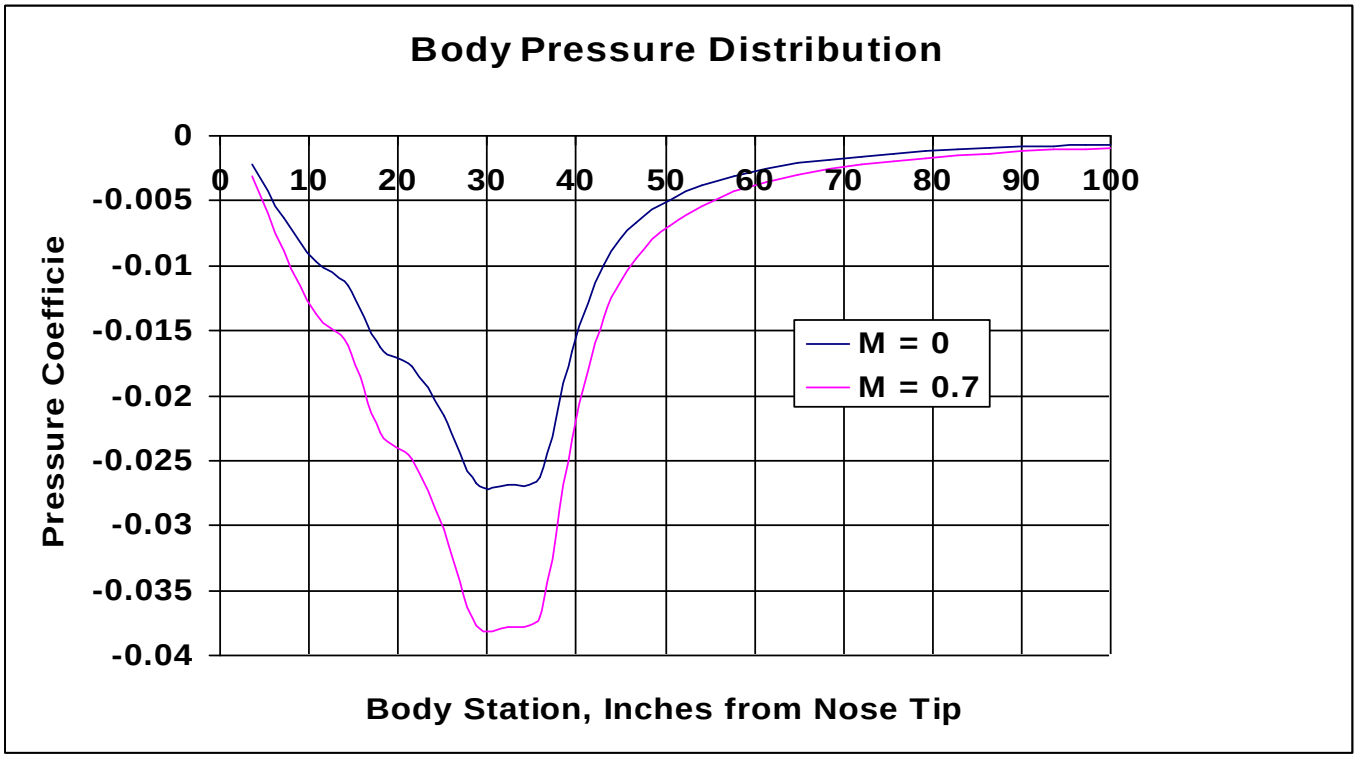


Fig. 2 Optimum Pressure Coefficients for L/D = 6

We see that the effect of compressibility is to reduce the minimum pressure coefficient.

Critical Mach Number

The crucial question is whether the Cp plotted in fig. 2 corresponds to supersonic flow or not. This is equivalent to asking what the critical Mach number for this L/D = 6 shape is; the critical Mach number is that lowest free stream Mach number for which the local surface Mach number is somewhere unity.

To find the critical Mach number, first, recall that the pressure coefficient is

$$Cp = (p - p\infty\,) \,/\, (\tfrac{1}{2}\, \gamma\, p\infty\, M\infty^2), \text{ where}$$

p = ambient pressure,
γ = specific heat ratio = 1.4 for dry air,
M = Mach number,
and the subscript ∞ refers to free stream conditions.
Thus,

$$Cp = (2 / (\gamma M\infty^2))* (p / p\infty – 1). \quad (2)$$

How do we find p / p∞ ? Recall that for one dimensional isentropic flow

$$p = p_T [1 + ½ (\gamma –1) M\infty^2] \wedge – \gamma / (\gamma –1), \text{ where} \quad (3)$$

p_T = the total, or stagnation, pressure, a constant along any stream tube absent shocks

Then, to find the pressure ratio needed in eq.(2) we can form the ratio of the surface pressure to the free stream pressure. If we further assume the local surface Mach number is unity, we find that

$$p / p\infty = [½ (\gamma +1) / [1 + ½ (\gamma –1) M\infty^2]] \wedge – \gamma / (\gamma –1). \quad (4)$$

Then, plug eq. (4) into eq. (2) and we can estimate the pressure coefficient leading to surface sonic flow. Note that this result will depend on free stream Mach number.

As you will see, the tool we have is basically Munk theory for incompressible flow over a body of revolution...Munk originally developed these ideas in the early 1920s to analyze the flow around airships, a big (no pun) deal in those days. There are two attachments, one of which is an Excel file in which the numbers are crunched, and the other a description of what's going on. The minimum pressure gradient forebody shape in the .doc was developed by cut and try methods. It had an L/D = 6 simply because that's what we flew on Gold Rush I and II. For our paper, we'll need other forebodies with other L/D values.

The question naturally arises regarding the pressure distributions for higher subsonic Mach numbers. As I told you, this is done using the Karman-Tsien correction. I googled Karman-Tsien compressibility correction, and found what we need in the second entry called Linear Compressibility. This will let us make a plot of Cp vs. body station for any subsonic free stream Mach number so long as there are no shocks. Note that the starting point for this is an incompressible Cp = -0.27, and the knowledge that the maximum Mach number expected in flight is ~ 0.83.

References
1. Hoult, C.P., "Nose Pressure Distribution", RST memo, 06 July 2010
2. Anon., "Desktop Aeronautics" web site, Linear Compressibility